

Technical Requirements Document

Frontend Stack

- React
- React Flow
- JavaScript
- CSS Modules / Styled Components

Backend Stack

- Python
- FastAPI

Architecture

Frontend

React Components ↓ React Flow Canvas ↓ Node System ↓ Submit API

Backend

FastAPI ↓ Pipeline Parser ↓ Graph Validator

Node Architecture

BaseNode Component

Reusable component responsible for:

- Node container
- Labels
- Inputs
- Outputs
- Shared styling
- Handle rendering

All nodes inherit configuration from BaseNode.

New Node Types

1. API Node

2. Condition Node
3. Database Node
4. Math Node
5. Notification Node

Purpose is to demonstrate abstraction flexibility.

Dynamic Variable Parsing

Regex:

```
/{{\s*([a-zA-Z_$][a-zA-Z0-9_$]*)\s*}}/g
```

Output:

```
{{name}}  
{{inputData}}  
{{user_email}}
```

Creates dynamic handles.

Backend API

POST

/api/pipelines/parse

Request

```
{  
  "nodes": [],  
  "edges": []  
}
```

Response

```
{  
  "num_nodes": 10,  
  "num_edges": 12,  
}
```

```
"is_dag": true  
}
```

DAG Detection

Approach:

Topological Sort

or

DFS Cycle Detection

Time Complexity:

$O(V + E)$

Performance Requirements

- Instant node rendering
 - Dynamic handle updates under 100ms
 - DAG validation under 500ms
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Security

- Input validation
 - JSON schema validation
 - Prevent malformed requests
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Deployment

Frontend:

- Vercel / Netlify

Backend:

- Render / Railway / Local FastAPI